COURSE SYLLABUS:

Field of study: **Computer Science**	Profile: **practical**	Level of studies: **first-cycle studies**	Year of study: **II**	Language of instruction: **English**
Academic year: **2025/2026** **Semester (winter/summer): summer**	Form (full-time / part-time): **full-time**	Status (compulsory / elective): **compulsory**	Prerequisites: **In accordance with the rules of studying at the Faculty of Design, par. 11. sec. 6-8. (Rules for qualification for the next year of study).**	
Number of hours in the study plan: **48**	Number of ECTS points: **5**	Total number of student work hours (1 ECTS=25-30h of work): **125**		

Course coordinator

Name and surname, academic degree:
Agnieszka Zbrzezny, PhD

Course components, form of classes, number of hours, instructor

Tyoe of classes	*Number of hours*	*Instructor*
Lectures	*24*	**Agnieszka Zbrzezny, Ph.D.**
Laboratories	*24*	**Krzysztof Ropiak, Ph.D. eng.**

REFERENCE TO FIELD-SPECIFIC LEARNING OUTCOMES (symbols of field-specific outcomes)	LEARNING OUTCOMES FOR THE COURSE	TASKS TO COMPLETE
INF1_W03 INF1_W05 INF1_W10	At an advanced level, the student knows and understands the principles of designing and analysing data processing and analysis algorithms and knows how to apply them to publicly available data repositories. The student knows and understands the structure and principles of managing data repositories and their potential use in complex projects, data processing, and data analytics. The student knows and understands selected artificial intelligence techniques and methods used in data analysis.	test practical tasks
INF1_U03	Is able to select and use IT tools and programming libraries to support the process of data analysis and processing.	test practical tasks
INF1_K01 INF1_K02	Knows that data analysis and processing are areas that require continuous training	practical tasks

Tasks to complete - characteristics	Implementation deadline	Scope of material to be prepared by students	Number of points to be scored	
			Max	Min
test	within 2 weeks of the end of classes	*all material from all classes*	50	26
practical tasks	carried out continuously in the laboratory	*material related to the issues discussed in previous lectures*	50	26

NOTE!

The student has the right to one retake of each task not passed on the first attempt. There is no possibility to improve passed tasks.

Attendance at classes is mandatory. A maximum of 2 unexcused absences are allowed. In case of an excused additional absence, the student will receive an additional task from the instructor.

CONVERTING POINTS TO GRADE:

PASSING THE COURSE requires the student to obtain **more than 50% of the points possible to obtain for each task** described in the table above. The final grade, provided that >50% of points are obtained for each task, results from the sum of points obtained. The grading scale is as follows:

GRADING SCALE:
52 – 60 – means 3 (satisfactory)
61 – 70 – means 3+ (satisfactory plus)
71 – 80 – means 4 (good)
81 – 90 – means 4+ (good plus)
91 – 100 – means 5 (very good)

COURSE CONTENT	
TYPE of classes: LECTURE	
Topics covered in classes:	1. Basics of Statistical Data Analysis 2. Data Visualization 3. Data Acquisition, Processing, and Storage 4. Data Cleansing 5. Regression Analysis
TYPE of classes: laboratory	
Topics covered in classes:	Practical, multi-faceted analysis of real-world datasets covering topics discussed in the lecture.

	1. Working in the Jupyter environment 2. NumPy and Pandas 3. Statistics and linear algebra using Python 4. Data visualization (Matplotlib, seaborn, bokeh) 5. Data acquisition, processing, and storage 6. Data cleaning 7. Regression

Required Literature:

Wes McKinney. Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter. O'Reilly Media, 2022

Siegmund Brandt. Data Analysis: Statistical and Computational Methods for Scientists and Engineers

Supplementary Literature:

Peter Bruce. Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python, O'Reilly Media, 2020

Jake VanderPlas. Python Data Science Handbook. 2nd Edition. O'Reilly Media, 2022